

Mini_Project4

October 3, 2026

```
[1]: pip install pandas
```

```
Requirement already satisfied: pandas in
C:\Users\surface\AppData\Roaming\Python\Python314\site-packages (2.3.3)
Requirement already satisfied: numpy>=1.26.0 in
C:\Users\surface\AppData\Roaming\Python\Python314\site-packages (from pandas)
(2.5.3)
Requirement already satisfied: python-dateutil>=2.8.2 in
C:\Users\surface\AppData\Roaming\Python\Python314\site-packages (from pandas)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
C:\Users\surface\AppData\Roaming\Python\Python314\site-packages (from pandas)
(2026.4)
Requirement already satisfied: tzdata>=2022.7 in
C:\Users\surface\AppData\Roaming\Python\Python314\site-packages (from pandas)
(2026.4)
Requirement already satisfied: six>=1.5 in
C:\Users\surface\AppData\Roaming\Python\Python314\site-packages (from python-
dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```
[2]: import pandas as pd
```

```
[3]: data=pd.read_csv(r"C:\Users\surface\Downloads\Salary_Data.csv")
```

```
[4]: data.head()
```

```
[4]:   YearsExperience  Salary
0             1.1  39343.0
1             1.3  46205.0
2             1.5  37731.0
3             2.0  43525.0
4             2.2  39891.0
```

```
[5]: data.tail()
```

```
[5]:   YearsExperience  Salary
25             9.0  105582.0
```

26	9.5	116969.0
27	9.6	112635.0
28	10.3	122391.0
29	10.5	121872.0

```
[6]: data.info
```

```
[6]: <bound method DataFrame.info of      YearsExperience      Salary
0          1.1    39343.0
1          1.3    46205.0
2          1.5    37731.0
3          2.0    43525.0
4          2.2    39891.0
5          2.9    56642.0
6          3.0    60150.0
7          3.2    54445.0
8          3.2    64445.0
9          3.7    57189.0
10         3.9    63218.0
11         4.0    55794.0
12         4.0    56957.0
13         4.1    57081.0
14         4.5    61111.0
15         4.9    67938.0
16         5.1    66029.0
17         5.3    83088.0
18         5.9    81363.0
19         6.0    93940.0
20         6.8    91738.0
21         7.1    98273.0
22         7.9   101302.0
23         8.2   113812.0
24         8.7   109431.0
25         9.0   105582.0
26         9.5   116969.0
27         9.6   112635.0
28        10.3   122391.0
29        10.5   121872.0>
```

```
[7]: data.describe()
```

```
[7]:      YearsExperience      Salary
count      30.000000      30.000000
mean         5.313333    76003.000000
std          2.837888    27414.429785
min          1.100000    37731.000000
25%          3.200000    56720.750000
```

50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

```
[8]: data.shape
```

```
[8]: (30, 2)
```

```
[9]: data.columns
```

```
[9]: Index(['YearsExperience', 'Salary'], dtype='object')
```

```
[10]: data.dtypes
```

```
[10]: YearsExperience    float64
Salary                float64
dtype: object
```

```
[11]: data.isnull().sum()
```

```
[11]: YearsExperience    0
Salary                0
dtype: int64
```

```
[12]: int(data.duplicated().sum())
```

```
[12]: 0
```

```
[13]: data.drop_duplicates(inplace=True)
```

```
[14]: data.nunique()
```

```
[14]: YearsExperience    28
Salary                30
dtype: int64
```

```
[15]: x=data[['YearsExperience']]
y=data[['Salary']]
print(x)
print(y)
```

	YearsExperience
0	1.1
1	1.3
2	1.5
3	2.0
4	2.2
5	2.9
6	3.0

7	3.2
8	3.2
9	3.7
10	3.9
11	4.0
12	4.0
13	4.1
14	4.5
15	4.9
16	5.1
17	5.3
18	5.9
19	6.0
20	6.8
21	7.1
22	7.9
23	8.2
24	8.7
25	9.0
26	9.5
27	9.6
28	10.3
29	10.5

	Salary
0	39343.0
1	46205.0
2	37731.0
3	43525.0
4	39891.0
5	56642.0
6	60150.0
7	54445.0
8	64445.0
9	57189.0
10	63218.0
11	55794.0
12	56957.0
13	57081.0
14	61111.0
15	67938.0
16	66029.0
17	83088.0
18	81363.0
19	93940.0
20	91738.0
21	98273.0
22	101302.0
23	113812.0

```
24 109431.0
25 105582.0
26 116969.0
27 112635.0
28 122391.0
29 121872.0
```

```
[16]: from sklearn.model_selection import train_test_split
      x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,
      ↪random_state=12)
```

```
[22]: from sklearn.linear_model import LinearRegression
      slr=LinearRegression()
      slr.fit(x_train,y_train)
      slr.predict(x_test)
```

```
[22]: array([[ 55675.5913984 ],
             [100372.57413579],
             [ 81352.58148158],
             [ 68038.58662364],
             [ 37606.59837691],
             [ 92764.5770741 ]])
```

```
[23]: from sklearn.metrics import r2_score
      y_pred=slr.predict(x_test)
      r2=r2_score(y_test,y_pred)
      print(r2)
```

```
0.9429269163491423
```